

TIC 388939913

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 5.541189 d (ALIASED, see below)
-> corrected period: 11.082379 d
Epoch (t0) 2554.9692 BTJD
Duration 0.1200 d
Depth 7.900 % (78,999 ppm)
SNR 154.2

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

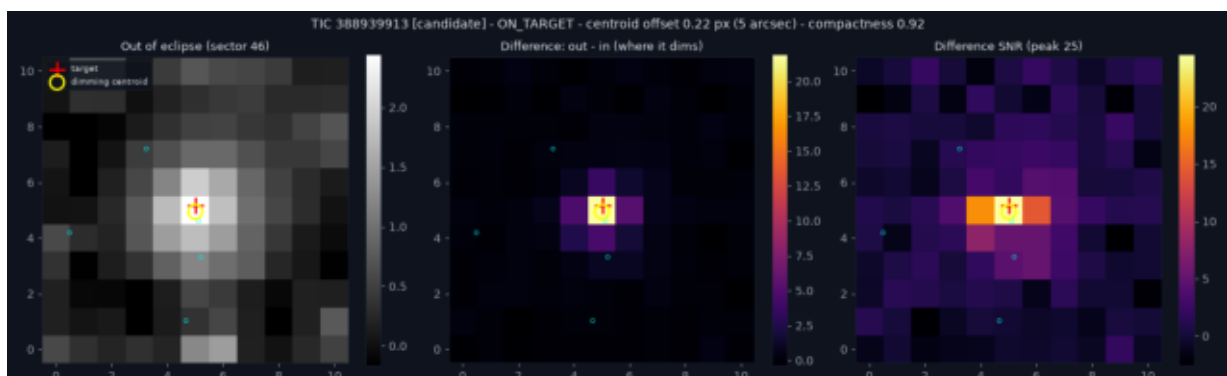
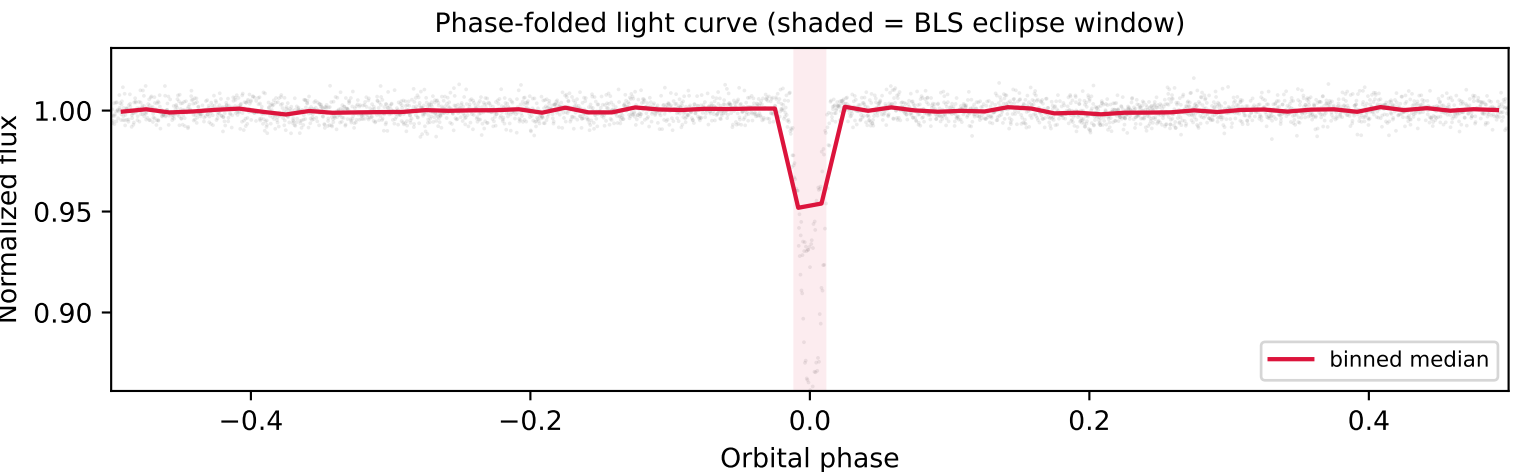
INDEPENDENT CHECKS (f3i gates)

Eclipses observed 4 (gate ≥ 3)
Robust depth ratio 0.87 (gate ≥ 0.50)
Robust depth sigma 105.9 (gate ≥ 7.0)
Odd/even z-score 57.51 (ALIAS FLAGGED)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 46 light-curve dip in this sector: 98.2 sigma t0 refined by 0.0025 phase
Difference-image peak SNR 24.54 (gate ≥ 5.0) Compactness 0.92 (gate ≥ 0.50)
Dimming centroid offset 0.22 px (5 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: none within the cutout



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 57.51 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 11.082379 d, in any write-up.